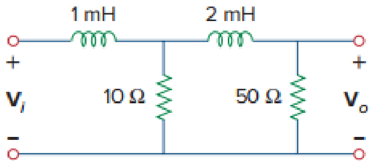
Chapter 9, Problem 14PP

Problem

Refer to the *RL* circuit in Fig. 9.36. If 10 V is applied to the input, find the magnitude and the phase shift produced at 5 kHz. Specify whether the phase shift is leading or lagging.

Figure 9.36:



Step-by-step solution

Step 1 of 6

Calculate the value of the angular frequency ω.

Write the expression for the angular frequency ω.

$$\begin{aligned}\omega &= 2\pi f \\ &= 2(3.14)\left(5\times10^{3}\right) \\ &= 31415.9 \text{ rad/s}\end{aligned}$$

Therefore, the value of the angular frequency ω is 31415.9 rad/s.

Consider $\mathbf{Z}_1$ is the impedance of the 1 mH inductor.

$\mathbf{Z}_2$ is the impedance of the resistor $10\ \Omega$.

$\mathbf{Z}_3$ is the impedance of the 2 mH inductor in series with the resistor $50\ \Omega$.

Redraw the Figure 1 as shown in Figure 2 for finding $\mathbf{Z}_{\text{in}}$ at $\omega = 31415.9 \text{ rad/s}$.

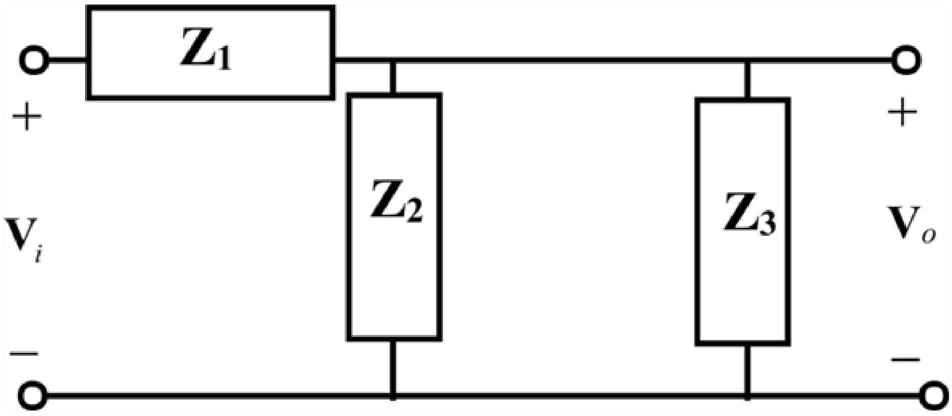


Figure 2

Step 2 of 6

Calculate the value of $\mathbf{Z}_1$.

Write the expression for $\mathbf{Z}_1$.

$$\mathbf{Z}_1 = j\omega L$$

Substitute 31415.9 rad/s for ω, 1×10^{-3} for L in this expression.

$$\begin{aligned}\mathbf{Z}_1 &= j(31415.9)1\times10^{-3} \\ &= j31.42\end{aligned}$$

Therefore, the value of $\mathbf{Z}_1$ is $j31.42\ \Omega$.

Step 3 of 6

Calculate the value of $\mathbf{Z}_2$.

Write the expression for $\mathbf{Z}_2$.

$$\mathbf{Z}_2 = 10\ \Omega$$

Therefore, the value of $\mathbf{Z}_2$ is $10\ \Omega$.

Step 4 of 6

Calculate the value of $\mathbf{Z}_3$.

Write the expression for $\mathbf{Z}_3$.

$$\mathbf{Z}_3 = j\omega L + 50$$

Substitute 31415.9 rad/s for ω, 2×10^{-3} for L in this expression.

$$\begin{aligned}\mathbf{Z}_3 &= j(31415.9)\left(2\times10^{-3}\right)+50 \\ &= j62.84+50 \\ &= (50+j62.84)\end{aligned}$$

Therefore, the value of $\mathbf{Z}_3$ is $(50+j62.84)\ \Omega$.

Step 5 of 6

Calculate the value of the impedance $\mathbf{Z}_2 \parallel \mathbf{Z}_3$.

Write the expression for the impedance $\mathbf{Z}_2 \parallel \mathbf{Z}_3$.

$$\begin{aligned}\mathbf{Z}_2 \parallel \mathbf{Z}_3 &= \frac{\mathbf{Z}_2\mathbf{Z}_3}{\mathbf{Z}_2+\mathbf{Z}_3} \\ &= \frac{10(50+j62.84)}{(10+50+j62.84)} \\ &= (9.205+j0.832)\end{aligned}$$

Therefore, the value of $\mathbf{Z}_2 \parallel \mathbf{Z}_3$ is $(9.205+j0.832)\ \Omega$.

Calculate the value of the output voltage $\mathbf{V}_{o1}$.

Apply voltage division rule across the impedance $\mathbf{Z}_2 \parallel \mathbf{Z}_3$ to find the output voltage $\mathbf{V}_{o1}$.

$$\begin{aligned}\mathbf{V}_{o1} &= \frac{\mathbf{Z}_2 \parallel \mathbf{Z}_3}{\mathbf{Z}_1+\mathbf{Z}_2 \parallel \mathbf{Z}_3}\mathbf{V}_i \\ &= \frac{(9.205+j0.832)}{(j31.42+9.205+j0.832)}(10) \\ &= \frac{(9.205+j0.832)}{(9.205+j32.252)}(10) \\ &= (0.99-j2.57)\end{aligned}$$

$$\mathbf{V}_{o1} = 2.755\angle -68.9^\circ\ \text{V}$$

Therefore, the value of $\mathbf{V}_{o1}$ is $2.755\angle -68.9^\circ\ \text{V}$.

Step 6 of 6

Calculate the value of the output voltage $\mathbf{V}_o$.

Apply voltage division rule across the resistor $50\ \Omega$ in $\mathbf{Z}_3$ to find the output voltage $\mathbf{V}_o$.

$$\begin{aligned}\mathbf{V}_o &= \frac{50}{(50+j62.84)}\mathbf{V}_{o1} \\ &= 0.622\angle -51.49^\circ(2.755\angle -68.9^\circ\ \text{V}) \\ &= (0.622\times 2.755)\angle -(51.49^\circ+68.9^\circ)\ \text{V} \\ &\cong 1.7161\angle -120.39^\circ\ \text{V}\end{aligned}$$

Therefore, the value of output voltage $\mathbf{V}_o$ is $1.7161\angle -120.39^\circ\ \text{V}$.

Hence, the magnitude and phase of the output voltage $\mathbf{V}_o$ is $\boxed{1.7161\ \text{V}}$ and $\boxed{\angle -120.39^\circ}$.

Clearly from the expression of the output voltage $\mathbf{V}_o$, the output voltage $\boxed{\text{lags}}$ the input voltage.